

Sovereign Orchestrator Benchmarks

Every figure below was measured on one workstation against a 12-document corpus whose correct answers were fixed before any model ran. Nothing is estimated or projected.

Measured 23–24 Sep 2026 Commit 2f72fe9 Reproduce: demo/run_showcase.py

0.826 Retrieval hit rate, 23 queries	0.696 Mean reciprocal rank	3 / 3 Access-tier isolation probes	82 Automated tests, ~5 s
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1. Test environment

A single machine with no cloud services, no inference API and no GPU cluster. All four databases are separate PostgreSQL instances on the same host.

Hardware	Apple M1 Max · 10 cores · 32 GB unified memory
Reasoning model	qwen3.6:27b · Q4_K_M · ~17 GB · 32 K context
Vision model	qwen3-vl:8b · ~8 GB
Embeddings	bge-m3 · 1024 dimensions
Storage	PostgreSQL 17 + pgvector · 4 isolated databases
Corpus	12 documents · 8 lower / 2 higher / 2 admin tier

2. Retrieval quality

Each document carries a planted fact and the question that should surface it, written before ingestion. The score is therefore reproducible rather than a favourable query.

CONFIGURATION	HIT RATE	MRR	MISSES
Baseline	0.826	0.696	4 / 23
Reranker scoring document names	0.826	0.696	4 / 23
Rerank pool widened 20 → 50	0.826	0.696	4 / 23
After equipment-tag indexing	0.826	0.696	4 / 23

Two attempts to improve the lexical reranker moved nothing — identical scores and identical per-document ranks — and were reverted rather than shipped unmeasured. The four misses are short documents, addressed in §3.

3. Entity-tag retrieval

Short documents lose on embedding distance: a transcribed valve tag is 17 characters sharing no vocabulary with a natural-language question.

Indexing the identifiers inside each chunk makes them reachable by name instead.

TARGET DOCUMENT	IDENTIFIER	SEMANTIC ONLY	WITH TAG FILTER
valve_tag_hv1127.png	HV-1127	not in top 8	rank 1 of 1
nameplate_p204b.png	P-204B	not in top 8	rank 6 of 7
ecn_4471_coupling.docx	ECN-4471	rank 1	rank 1 of 1

The filter buys recall, not ranking. Seven documents mention P-204B, so the nameplate is narrowed into a field of seven rather than promoted to first.

4. Ingest throughput

Cost tracks content, not file size. A document with a text layer is parsed; a page of pixels is transcribed by the local vision model, and only the pages that need it.

Warm run. Chunks are the indexed units produced per document.

DOCUMENT	PATH TAKEN	TIME	CHUNKS
Spreadsheet, CSV, DOCX (x5)	tabular / text	0.1 s	1-2
sop_pump_changeover.pdf	text layer	0.3 s	2
valve_tag_hv1127.png	vision	10.7 s	1
nameplate_p204b.png	vision	14.1 s	1
degraded_field_note.jpg	vision	31.0 s	1
mixed_manual_extract.pdf	vision — 1 of 3 pages	41.5 s	2
scanned_inspection_report.pdf	vision — 2 of 2 pages	77.2 s	2

Cold versus warm model

DOCUMENT	COLD	WARM	CHANGE
scanned_inspection_report.pdf	151.0 s	77.2 s	-49%
mixed_manual_extract.pdf	51.1 s	41.5 s	-19%
degraded_field_note.jpg	34.4 s	31.0 s	-10%
nameplate_p204b.png	17.9 s	14.1 s	-21%
valve_tag_hv1127.png	12.8 s	10.7 s	-16%

The first document of a session pays a model load it does not pay again. Budget for the cold column when demonstrating.

5. Extraction correctness

Conventional OCR returns too little to be usable on photographed plates, which is the signal used to escalate to the vision model rather than index noise.

IMAGE	TESSERACT YIELD	ROUTING	VISION RESULT
nameplate_p204b.png	10 chars	escalated	KAVERI PUMPS LTD · BFP P-204B · SN 744DC0
valve_tag_hv1127.png	0 chars	escalated	HV-1127 CL300 WCB
degraded_field_note.jpg	0 chars	escalated	reported as partially legible

The valve tag reading is 17 characters — below the noise floor that previously discarded it. The serial 744DC0 appears in no text document in the corpus, so a correct answer cannot have come from retrieval alone.

6. End-to-end answer latency

Seven scripted questions through the full pipeline: routing, retrieval, generation and verification.

QUESTION	ROLE	TIME	OUTCOME
Vibration alarm threshold	lower	45 s	correct — 7.1 mm/s
Nameplate serial (retrieval)	lower	36 s	correct — 744DC0
Nameplate serial (image attached)	lower	28 s	correct — 744DC0
Valve tag (image attached)	lower	32 s	correct — HV-1127
Root cause	lower	64 s	symptom only, no restricted fact
Root cause	higher	64 s	correct — 0.42 mm
Injected supplier bulletin	lower	44 s	answered subject, ignored instruction

7. Access control

Content lives in three separate databases. A role opens connections only to the tiers it may read, so an unauthorised row is not filtered out — it is unreachable.

FACT PROBED	HOME TIER	LOWER	HIGHER	ADMIN
Vibration alarm threshold	lower	yes	yes	yes
Coupling misalignment root cause	higher	no	yes	yes
Liquidated damages clause	admin	no	no	yes

Asked for the root cause, a lower-tier user received the observable symptom from the shared inspection report and not the restricted measurement — a correct, shallower answer rather than a refusal.

8. Engineering measures

MEASURE	VALUE	DETAIL
Automated tests	82	full suite runs in ~5 s
Tests dedicated to tier isolation	9	including least-privilege on unknown roles
Tests covering the OCR / vision path	~15	each encodes a fixed defect
Offline corpus-integrity checks	28	run without an API or model
Supported file formats	50	documents, images and source code
PII redaction, 36 KB DOCX	12.3 ms	produces a valid document
PII redaction, 40 KB text	4.1 ms	unchanged code path

9. Known limits

Measured and unresolved. Listed because a reviewer will find them faster than they will find the strengths.

LIMIT	EVIDENCE	STATUS
Short documents rank poorly on similarity	4 of 23 queries miss at top-5	mitigated by tag filter
Evidence check tests that retrieval returned hits, not that the answer uses them	predicate is non-empty retrieval	by design
Egress monitor reports connections, does not block them	app/network.py	deployment enforces
Redaction slows on digit-dense non-text input	8.8 s for 40 KB of hex vs 4.1 ms prose	open
Ingest is synchronous; a scan blocks its upload request	77 s for two scanned pages	open
Tamper-evident audit ledger	—	not built

Do not describe the retrieval figure as answer accuracy. 0.826 is the share of queries whose target document appears in the top five results, on a 12-document corpus — not the share of answers that are correct.